THE 0-SCHUR–WEYL TENSOR SPACE HAS NO $L_I \boxtimes M_I$ FILTRATION

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For Clio (cc Robin).
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Summary. Let $V = \mathbb{C}^n$ with $n \geq 2$, and let $H = H_N(0)$ act on $V^{\otimes N}$ by Jung–Oh (arXiv:2509.26194, (4.5)). Put $S_0 = \mathrm{End}_H(V^{\otimes N})$. Then $V^{\otimes N}$ has an (S_0, H)-bimodule filtration with layers $L_I \boxtimes M_I$ *if and only if* $N \leq 2$. So the Krob–Thibon K_0 identity $\sum_I F_I R_I$ does not lift to a filtration.
Grades: Theorem 1 and Theorem′ proved (internal grade), not externally checked. The direction ($\Leftarrow$) cites the labelling of Jung–Oh Thm 4.13. Addendum (the U_0 variant): computed, not proved.

1. Setting and statement

$V = \mathbb{C}^n$, $n \geq 2$. $H = H_N(0)$ acts on the right of $V^{\otimes N}$ by Jung–Oh (4.5), with $\bar{\pi}_i^2 = -\bar{\pi}_i$. On a word $w = w_1 \cdots w_N$ with $a = w_i$, $b = w_{i+1}$:

$$w \cdot \bar{\pi}_i = \begin{cases} s_i w \text{ (swap)} & a < b, \\ -w & a > b, \\ 0 & a = b. \end{cases}$$

$S_0 = \mathrm{End}_H(V^{\otimes N})$. M_I denotes the indecomposable projective H-modules (I a composition with $\ell(I) \leq n$), and L_I the simple S_0-modules.

Theorem 1. *Let $n \geq 2$. $V^{\otimes N}$ has a filtration by (S_0, H)-sub-bimodules with subquotients $L_I \boxtimes M_I$ (each admissible I once) if and only if $N \leq 2$.*

The ($\Rightarrow$) direction proves something much stronger.

Theorem′ (strong form)**.** *Let $n \geq 2$ and $N \geq 3$. There is no chain of S_0-stable H-submodules $0 = X_0 \subset X_1 \subset \cdots \subset X_m = V^{\otimes N}$ in which every subquotient X_k/X_{k-1} is, as an H-module, isomorphic to $P_k^{\oplus d_k}$ for a single indecomposable projective P_k.*

The S_0-structure of the layers plays no role: neither simplicity of L_I nor its character F_I is used. As an H-module, $L \boxtimes M_I \cong M_I^{\oplus \dim L}$, so Theorem′ implies ($\Rightarrow$).

2. Proof

Lemma 2 (stable summands are Hom-orthogonal)**.** *Let H be any finite-dimensional algebra, V a finite-dimensional right H-module and $S = \mathrm{End}_H(V)$. Let $X \subset V$ be an S-stable H-submodule with V/X projective. Then $V = X \oplus Y$ for some H-submodule Y, and $\mathrm{Hom}_H(X, Y) = 0$.*

Proof. V/X is projective, so $V \to V/X$ splits: $V = X \oplus Y$. Let $p : V \to X$ be the projection along Y; it is H-linear. Given $f \in \mathrm{Hom}_H(X, Y)$, set $F = f \circ p \in S$. Then $F(X) = f(X) \subset Y$, and $F(X) \subset X$ by S-stability. Hence $f(X) \subset X \cap Y = 0$. □

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Lemma 2 needs S to be the *full* commutant, since p must lie in S. This is the only place $S_0 = \mathrm{End}_H(V)$ is used.

Lemma 3 (two hook summands)**.** *Let $N \geq 3$ and fix two letters $0 < 1$ of V. Let u_p be the word with a single 1 in position p (other letters 0), and v_p the word with a single 0 in position p (other letters 1). Put $W = \mathrm{span}\{u_1, \dots, u_{N-1}\}$ and $W' = \mathrm{span}\{v_2, \dots, v_N\}$.*

(a) W and W' are H-direct summands of $V^{\otimes N}$.
(b) $f : W \to W'$, $u_{N-1} \mapsto v_N$, $u_k \mapsto 0$ $(k < N-1)$, is a nonzero H-map.
(c) $g : W' \to W$, $v_2 \mapsto u_1$, $v_p \mapsto 0$ $(p \geq 3)$, is a nonzero H-map.
(d) $W \not\cong W'$.
(e) W and W' are local (hence indecomposable), with simple tops $\mathbb{C}_{\{N-1\}}$ and $\mathbb{C}_{\{1\}}$.

Proof. From the action: $u_p\bar{\pi}_{p-1} = u_{p-1}$, $u_p\bar{\pi}_p = -u_p$, $u_p\bar{\pi}_i = 0$ otherwise; $v_p\bar{\pi}_{p-1} = -v_p$, $v_p\bar{\pi}_p = v_{p+1}$, $v_p\bar{\pi}_i = 0$ otherwise.

(a) Let $U = \mathrm{span}\{u_p\}$, an H-stable weight space. W is stable (the $\bar{\pi}_i$ only move the 1 leftward or scale), $\sum_p u_p$ is killed by every $\bar{\pi}_i$ (as $(\sum u)\bar{\pi}_i = u_i - u_i = 0$), $\sum u \notin W$, and dimensions add, so $U = \mathbb{C}\cdot\sum_p u_p \oplus W$. Likewise $U' = \mathbb{C}\cdot\sum_p v_p \oplus W'$. Every weight space is an H-direct summand of $V^{\otimes N}$, since the action preserves letter content.

(b) $\mathrm{span}\{u_1, \dots, u_{N-2}\}$ is a submodule, so the kernel is stable. $u_{N-1}\bar{\pi}_{N-1} = -u_{N-1}$ and $v_N\bar{\pi}_{N-1} = -v_N$; $u_{N-1}\bar{\pi}_{N-2} = u_{N-2} \mapsto 0$ and $v_N\bar{\pi}_{N-2} = 0$ (letters $1, 1$); all other $\bar{\pi}_i$ kill both.

(c) $\mathrm{span}\{v_3, \dots, v_N\}$ is a submodule. $v_2\bar{\pi}_1 = -v_2$ and $u_1\bar{\pi}_1 = -u_1$; $v_2\bar{\pi}_2 = v_3 \mapsto 0$ and $u_1\bar{\pi}_2 = 0$ (letters $0, 0$, using $N \geq 3$); other $\bar{\pi}_i$ kill both.

(d) u_1 spans a submodule of W with $\bar{\pi}_1 = -1$, $\bar{\pi}_i = 0$ for $i \geq 2$. If $y = \sum_{p\geq 2} c_p v_p \in W'$ satisfies these relations, then $y\bar{\pi}_1 = -c_2 v_2 = -y$ forces $c_p = 0$ for $p \geq 3$, and then $y\bar{\pi}_2 = c_2 v_3 = 0$ forces $c_2 = 0$. So W' has no such submodule.

(e) W is generated by u_{N-1}. Let $R = \mathrm{span}\{u_1, \dots, u_{N-2}\}$, of codimension 1. If a submodule Z contains $z = c\,u_{N-1} + r$ with $r \in R$, $c \neq 0$, then since $u_k\bar{\pi}_{N-1} = 0$ for $k \leq N-2$, $z\cdot(-\bar{\pi}_{N-1}) = c\,u_{N-1}$, so $Z = W$. Thus R is the unique maximal submodule and W has simple top $\mathbb{C}_{\{N-1\}}$; in particular W is indecomposable. For W': it is generated by v_2, $v_p\bar{\pi}_1 = 0$ for $p \geq 3$, so $z\cdot(-\bar{\pi}_1) = c\,v_2$; the unique maximal submodule is $\mathrm{span}\{v_3, \dots, v_N\}$ and the top is $\mathbb{C}_{\{1\}}$. (This reproves (d), since the tops differ.) □

Proof of Theorem′. Suppose such a chain exists.

(1) Each subquotient is projective, so each V/X_k is projective and the chain splits as H-modules: $V \cong \bigoplus_k P_k^{\oplus d_k}$.
(2) By Krull–Schmidt and Lemma 3(a),(e), $W \cong P_a$ and $W' \cong P_b$ for some a, b; by (d), $a \neq b$.
(3) Let k be minimal with X_k containing a summand of type W or W'; say type W (the other case is symmetric, using g in place of f). Then $X_k \cong \bigoplus_{j\leq k} P_j^{d_j}$ has no summand of type W'.
(4) By Lemma 2, $V = X_k \oplus Y$, and by Krull–Schmidt the summand W' of V lies (up to isomorphism) in Y.
(5) Then $\mathrm{Hom}_H(X_k, Y) \supseteq \mathrm{Hom}_H(W, W') \ni f \neq 0$, contradicting Lemma 2. □

Proof of (⇐). $H_1(0) = \mathbb{C}$ and $H_2(0) = \mathbb{C}[\bar{\pi}]/(\bar{\pi}^2 + \bar{\pi}) \cong \mathbb{C} \times \mathbb{C}$ are semisimple. Then $V = \bigoplus_D \mathrm{Hom}_H(\mathbb{C}_D, V) \boxtimes \mathbb{C}_D$ and $S_0 = \prod_D \mathrm{End}(\mathrm{Hom}_H(\mathbb{C}_D, V))$, a direct sum of simple bimodules $L \boxtimes \mathbb{C}_D = L \boxtimes M_D$. For $n \geq 2$ both compositions of 2 occur. The labelling (character F_I) is cited from Jung–Oh Thm 4.13, not re-proved; for semisimple H any labelling works anyway. □

Verification. A script (`proofs/prove287/chk.py`, sympy, exact) checks for $N = 3, 4, 5, 6$, $n = 2$ letters: W, W' are submodules; $\dim \mathrm{Hom}(W, W') = \dim \mathrm{Hom}(W', W) = 1$; $\dim \mathrm{End}\, W = \dim \mathrm{End}\, W' = 1$. The Hom bases printed for $N = 3, 4, 5$ are exactly f and g.

3. Remarks and caveats

Remark 4 (the obstruction)**.** The obstruction is a 2-cycle $W \rightleftarrows W'$ in the Hom-quiver of the summands present: two *non-isomorphic* indecomposable projectives (hooks, tops at descent sets $\{N-1\}$ and $\{1\}$) with nonzero maps both ways. Iterating Lemma 2, an S-stable filtration by isotypic projective layers needs an ordering of the summand types in which Hom only goes "backwards"; a 2-cycle kills every ordering. (A general criterion along these lines is sketched in the proof record but is not needed here.)

Remark 5 (negative control: generic q)**.** At generic q the same weight spaces give $W \cong W'$ (both the Specht module $(N-1, 1)$, simple). Step (d) fails and the argument correctly says nothing.

Remark 6 (generality)**.** Theorem′ holds verbatim for any right $H_N(0)$-module V containing summands of both types W and W', with $S = \mathrm{End}_H(V)$.

Remark 7 (U_0 versus S_0)**.** The $U_0(\mathfrak{gl}_n) \times H$ version is a *different* question wherever $\rho(U_0) \subsetneq S_0$, as at $(n, N) = (2, 3)$ (Krob–Thibon, NCSF V, Note 5.7). Lemma 2 needs the full commutant, so the theorem above does not touch that version. See the Addendum for what has been computed.

Remark 8 (honest caveats)**.**

- The proof is elementary (Krull–Schmidt plus splitting of projective filtrations). The statement is plausibly folklore, e.g. in standardly-stratified language; no literature search for it has been done.
- Deng–Yang, JPAA 216 (2012) 1253 (Morita equivalence and self-injectivity of S_0) is *unread* and may bear on this.
- Grade: proved (internal grade), not externally checked.

4. Addendum: the U_0 variant (computed, not proved)

Grade: computed. Nothing in this section is proved for general (n, N).

Conventions. H-action as above. $U_0(\mathfrak{gl}_n)$ as in Jung–Oh Def. 4.2 with its coproduct $\Delta e_i = 1 \otimes e_i + e_i \otimes k_i$, $\Delta f_i = k_{i+1} \otimes f_i + f_i \otimes 1$, $\Delta k_i = k_i \otimes k_i$, and $\rho_V(e_i) = E_{i,i+1}$, $\rho_V(f_i) = E_{i+1,i}$, $\rho_V(k_i) = \sum_{j \neq i} E_{jj}$. The U_0 operators were checked to commute with $\rho(H)$.

Dimensions (exact over $\mathbb{Q}$):

(n, N)	$\dim \rho(U_0)$	$\dim S_0$
$(2, 3)$	19	20
$(2, 4)$	31	35
$(3, 3)$	165	165
$(2, 5)$	46	not computed

The $(2, 3)$ and $(2, 4)$ values agree with Krob–Thibon NCSF V Note 5.7.

Method ($n = 2$, exact sympy). Let $\varepsilon = \pi_{w_0}$ (with $\pi_i = 1 + \bar{\pi}_i$), a central idempotent of H cutting out the block of the simple projective $M_{(N)}$. Then $V^{\otimes N} = T \oplus Y$ with $T = V^{\otimes N}\varepsilon$, $Y = V^{\otimes N}(1 - \varepsilon)$, both stable under $A = \langle \rho(U_0), \rho(H) \rangle$. $\rho(U_0)|_T$ is the full matrix algebra, so $T = L_{(N)} \boxtimes M_{(N)}$ is simple and can be placed anywhere. $Y = \bigoplus_{a=1}^{N-1} P_{(a,N-a)}$, each type once with local endomorphism ring, so a bottom piece of type P_i is a graph $\{x + \varphi x\}$. The search solves the A-stability equations exactly, passes to quotients and recurses. On this decomposition

the weak (isotypic projective layers) and strong ($L \boxtimes M_I$) forms coincide. *Control:* dropping f_0 and/or e_0 from the generators, the search does find chains, so the solver is not vacuous.

Results.

(n, N)	$\dim \rho(U_0)\|_Y$ vs $\dim \mathrm{End}_H(Y)$	filtration?	reason
$(2, 3)$	3 vs 4	**NO**	no A-stable bottom piece of either type
$(2, 4)$	6 vs 10	**NO**	no A-stable bottom piece of any of 3 types
$(2, 5)$	10 vs 20	**NO**	no A-stable bottom piece of any of 4 types
$(3, 3)$	$\rho(U_0) = S_0$	**NO**	Theorem′ applies verbatim (no search run)

Mechanism at $(2, 3)$**.** With $W = P_{(2,1)}$ and $W' = P_{(1,2)}$: $\rho(U_0)|_Y = \mathrm{span}\{1_Y, e_0|_Y, f_0|_Y\}$. Here $f_0|_Y : W \to W'$ is nonzero and spans $\mathrm{Hom}_H(W, W')$ (the map f), and $e_0|_Y : W' \to W$ is nonzero and spans $\mathrm{Hom}_H(W', W)$ (the map g). The element of S_0 missing from $\rho(U_0)$ is the weight idempotent separating W from W', which Lemma 2 never needs. Directly: a bottom piece $X \cong W$ is a graph over W, so $f_0 X = f_0 W \subset W'$ is nonzero while $X \cap W' = 0$; symmetrically for W' using e_0. So the Hom 2-cycle $W \rightleftarrows W'$ is realised inside $\rho(U_0)$ itself.

Scope and doubts.

- The answer for general (n, N) is *open*. A plausible route (unverified): show f_i, e_i restricted to the hook summands are nonzero, so Lemma 2's argument runs with $\rho(U_0)$ in place of S_0.
- A different U_0 convention (e.g. Krob–Thibon's own coproduct) could give a different $\rho(U_0)$. The dimension match 19/20 with KT Note 5.7 suggests the images agree, but this was not checked.